8 **(a)** **(i)** *either* $P = V^2 / R$ *or* $I = 1200 / 230$ or 5.22 C1

$R = (230 \times 230) / 1200$

$R = 230^2 / 1200$ or $R = 230 / 5.22$ M1

$= 44.1\,\Omega$ $= 44.1\,\Omega$ A0 [2]

(ii) $R = \rho L / A$ C1

$= (1.7 \times 10^{-8} \times 9.2 \times 2) / (\pi \times \{0.45 \times 10^{-3}\}^2)$ M1

$= 0.492\,\Omega$ A0 [2]

(b) current $= 230 / 44.6$ C1

power $= (230 / 44.6)^2 \times 44.1$ C1

$= 1170\,\text{W}$ A1 [3]

(*allow full credit for solution based on potential divider*)

(c) e.g. less power dissipated in the heater / smaller p.d. across heater /

more power loss in cable / current lower B1

cable becomes heated / melts B1 [2]

(*any two sensible suggestions, 1 each, max 2*)

9 **(a)** nucleus emits α-particles or β-particles and/or γ-radiation B1